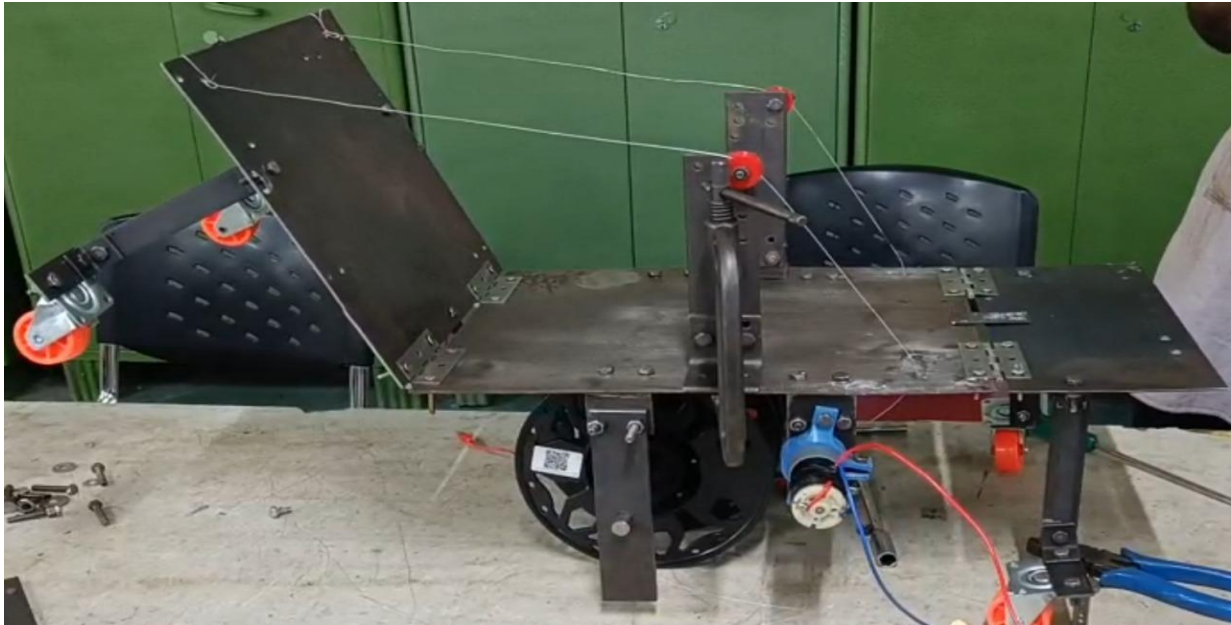


TA-212 LAB
PROJECT REPORT, 1st Semester 2025-26
AUTOMATIC
STRETCHER - WHEELCHAIR



Group No. – 13

Course Instructor: Dr. Sarvesh Mishra

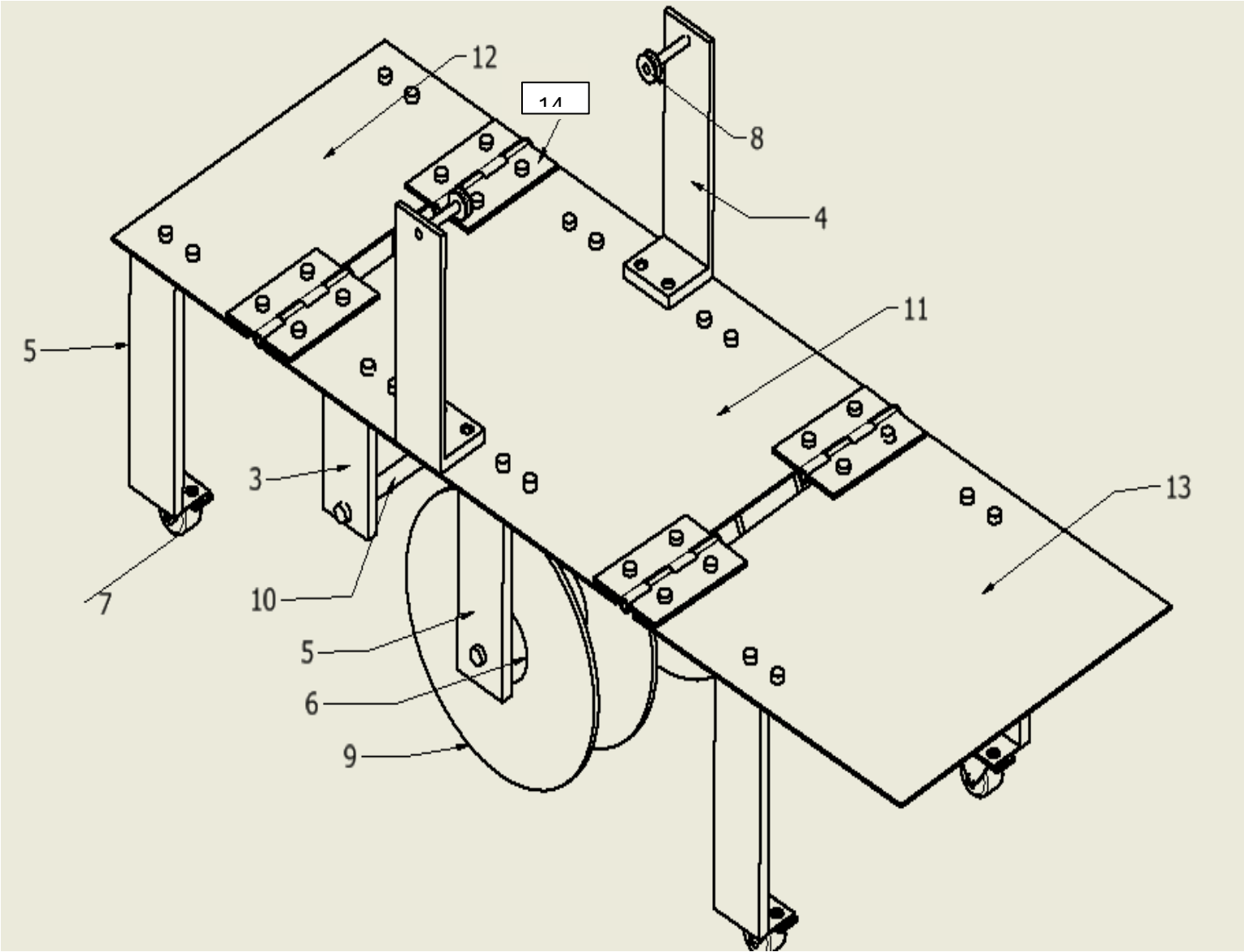
Tutor:

Lab In-Charge: Mr. Aman Singh

Guide: Mr. Dheeraj Soni

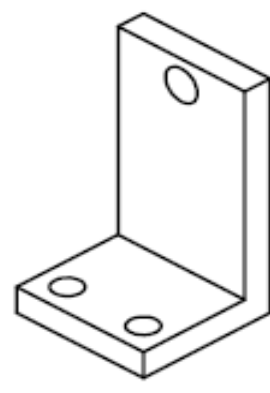
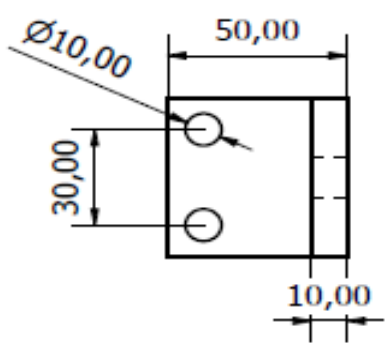
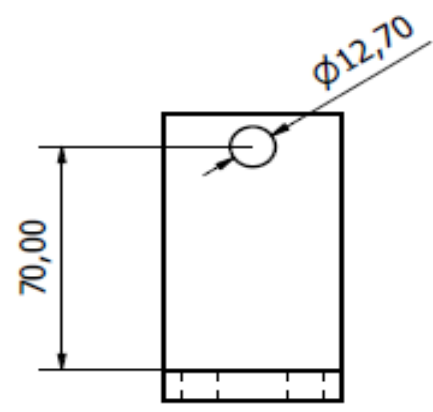
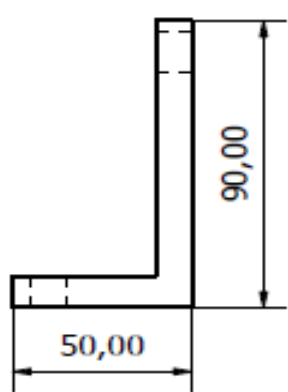
Group members:	Name	Roll No.
	1. Aashutosh Kushwaha	240022
	2. Uday Charan	240743
	3. Koppala Venkata Sai	240553
	4. Yash Kumar	241199
	5. Ishika Gupta	240475
	6. Aman Ahuja	240099
	7. Ritvija Katore	240878
	8. Shantanu Mohan	240961
	9. Rudra Sharma	240890

ISOMETRIC VIEW OF THE ASSEMBLY

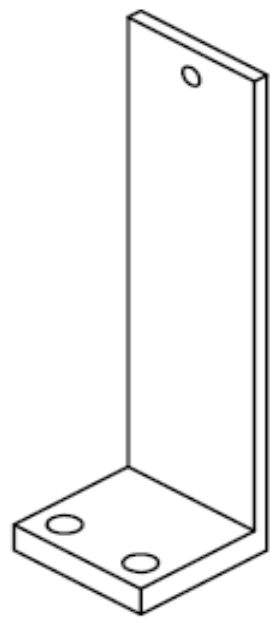
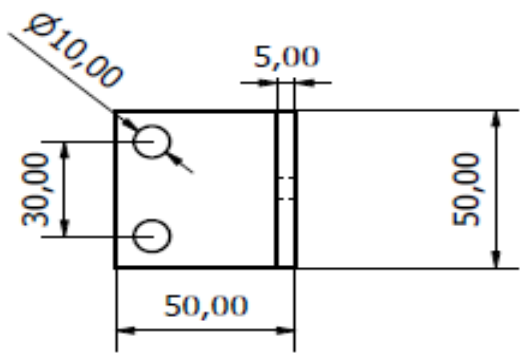
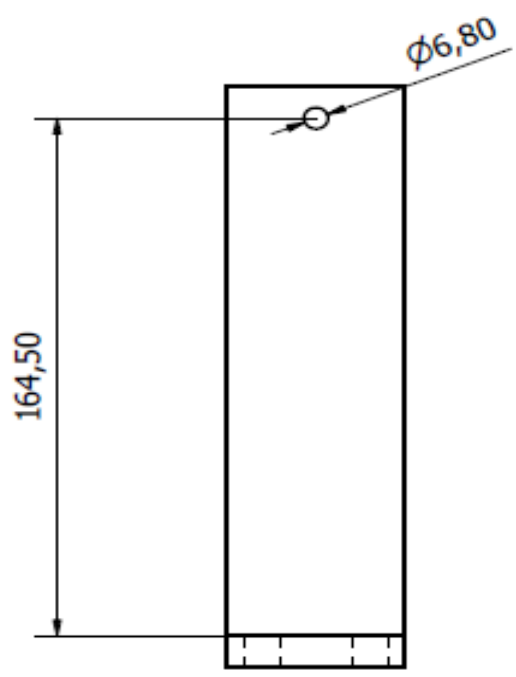
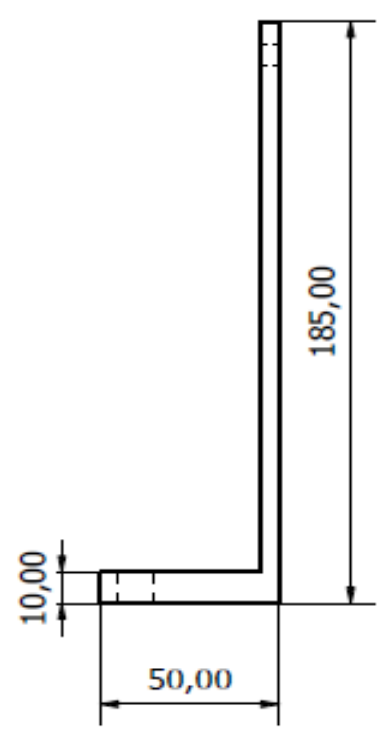


INDEX / PART LIST

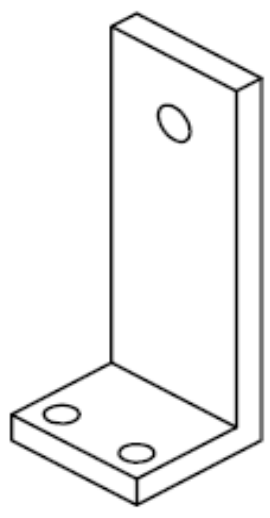
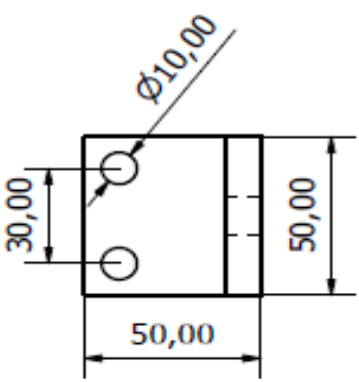
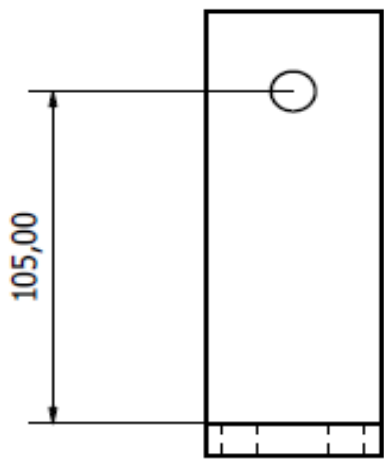
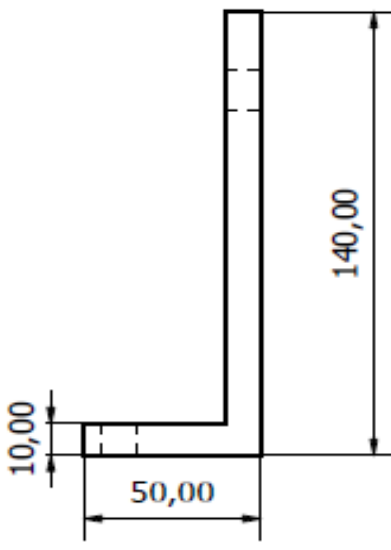
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1	Isometric drawing,/Manual Mixture	1	-	-	2
2	Index	1	-		3
3	Thread wound axle support	2	50x90x50	MS	4
4	Pulley Mounts	2	50x185x50	MS	5
5	Primary Wheel Mount	2	50x140x50	MS	6
6	Primary Wheel Shaft	2	Dia 60 x 90	MS	7
7	Castor Wheel	4	20x20x60	PLA	8
8	Pulley	2	Dia 20 x 7	MS	9
9	Primary Wheel (Spools)	2	Dia 210 x 70	PLA	10
10	Axle	2	Dia 12.7 x 315	MS	11
11	Base Plate	1	390 x 2 x 305	MS	12
12	Back Plate	1	150 x 2 x 305	MS	13
13	Front Plate	1	295 x 2 x 305	MS	14
14	Hinges	4	100 x 2 x 70	SS	15



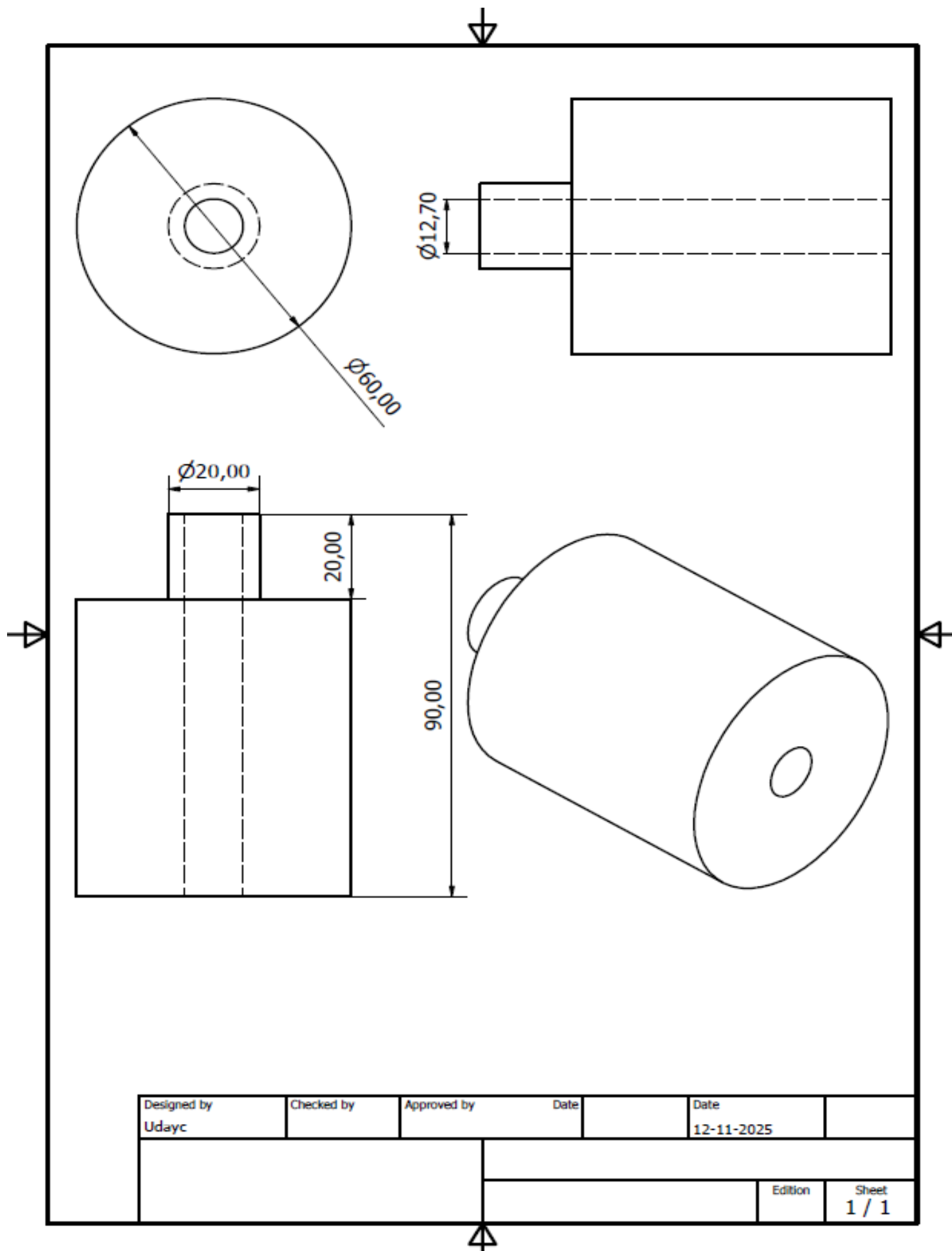
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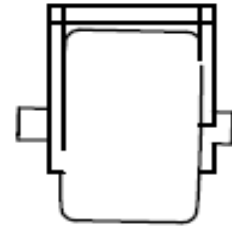
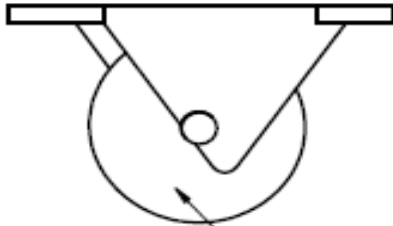
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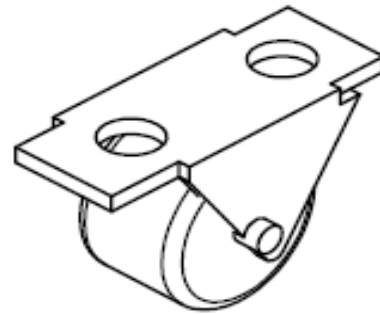
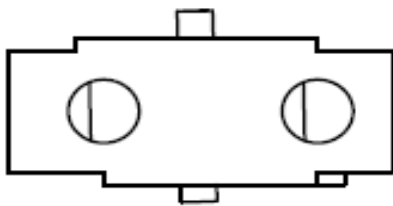
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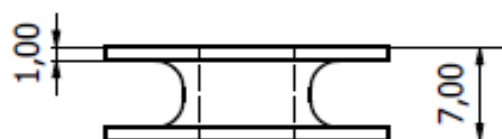
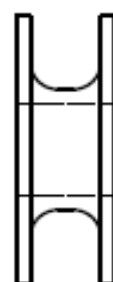
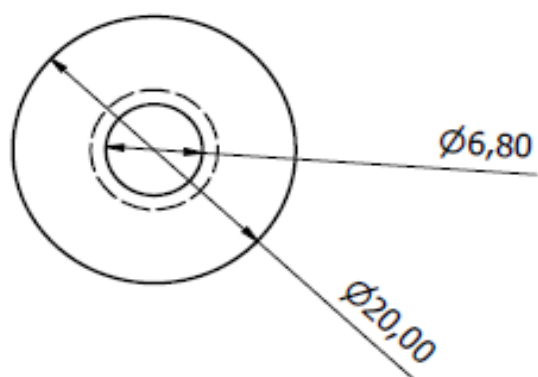


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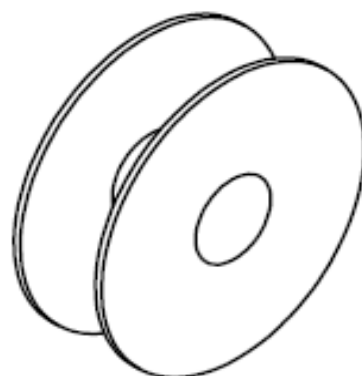
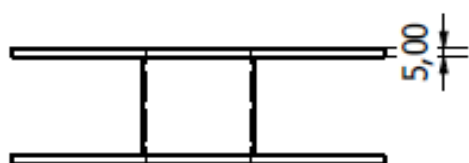
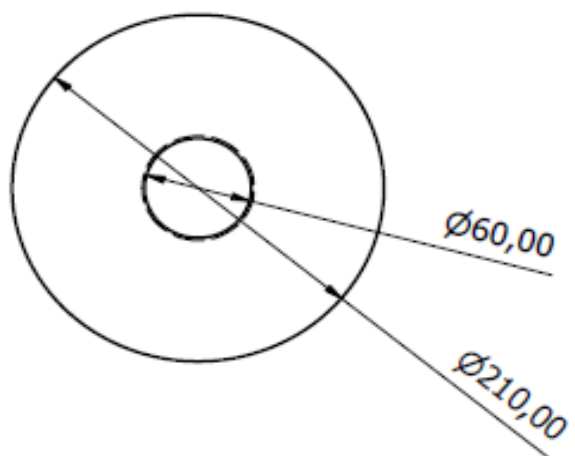


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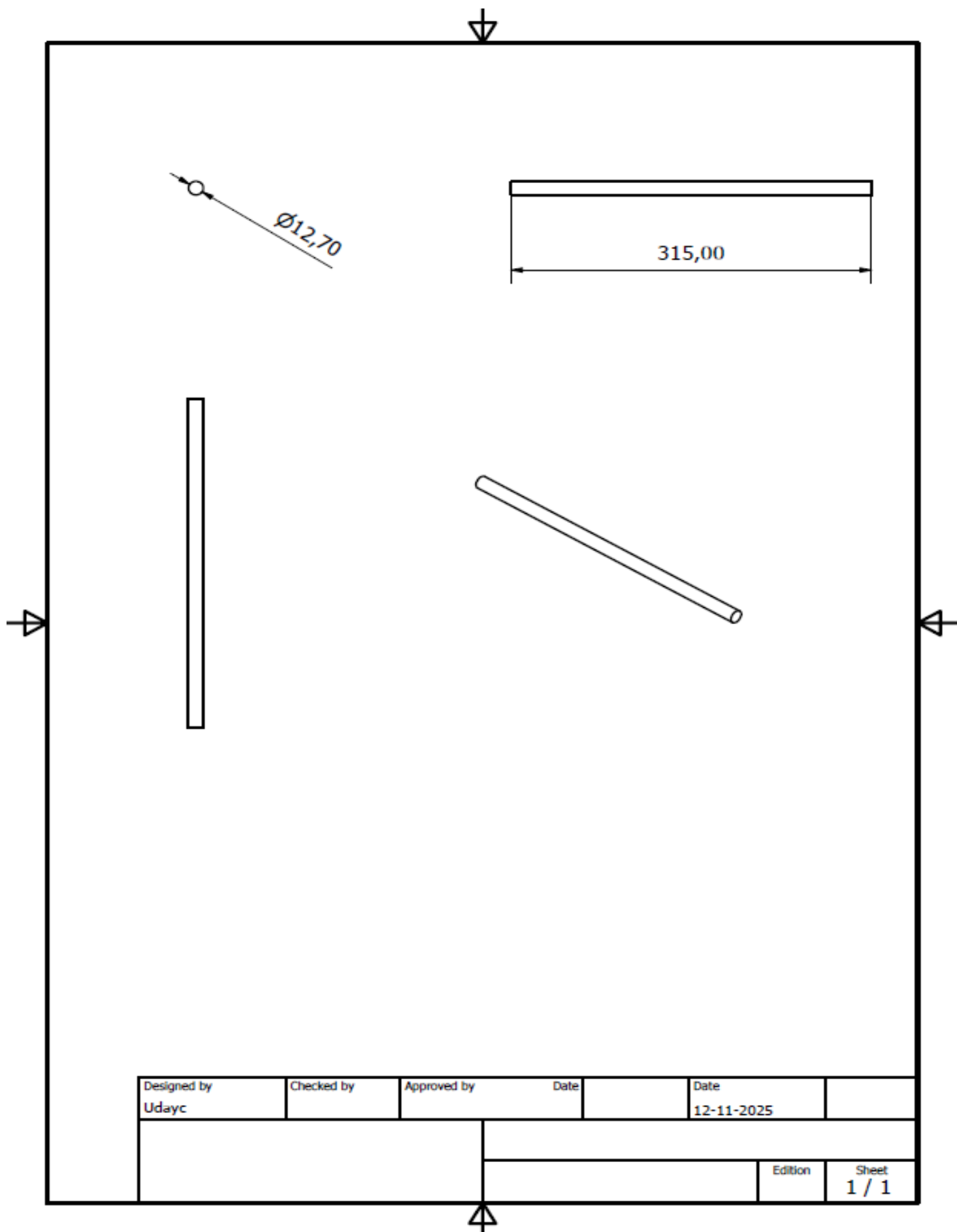


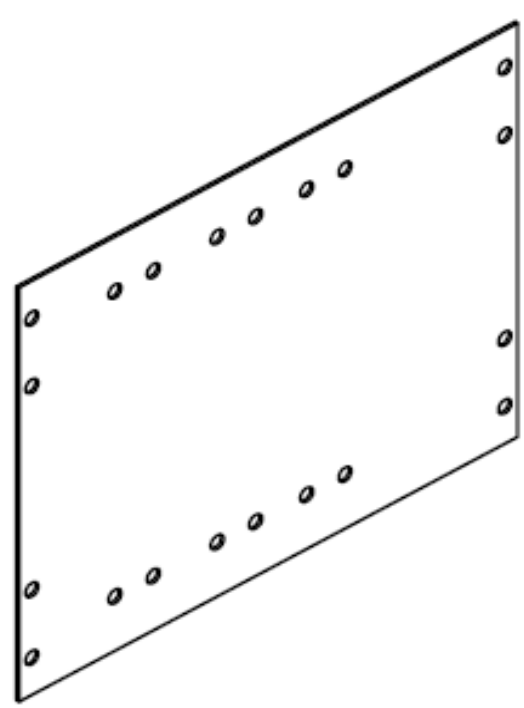
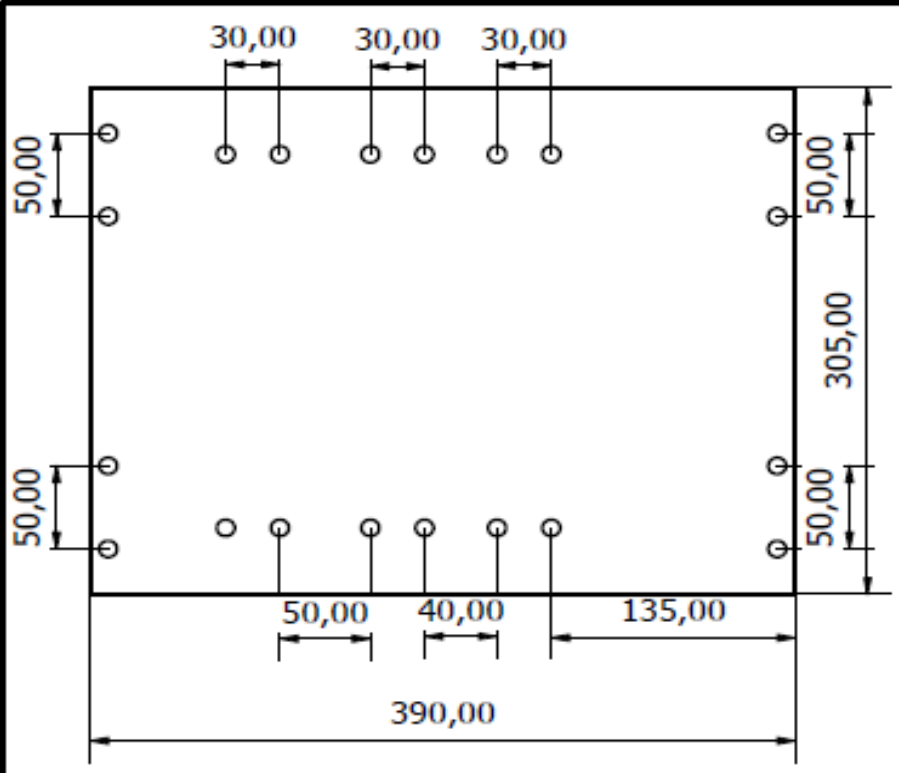


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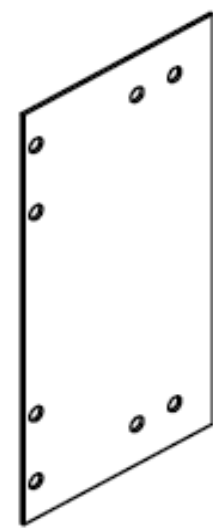
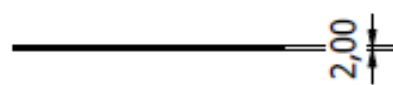
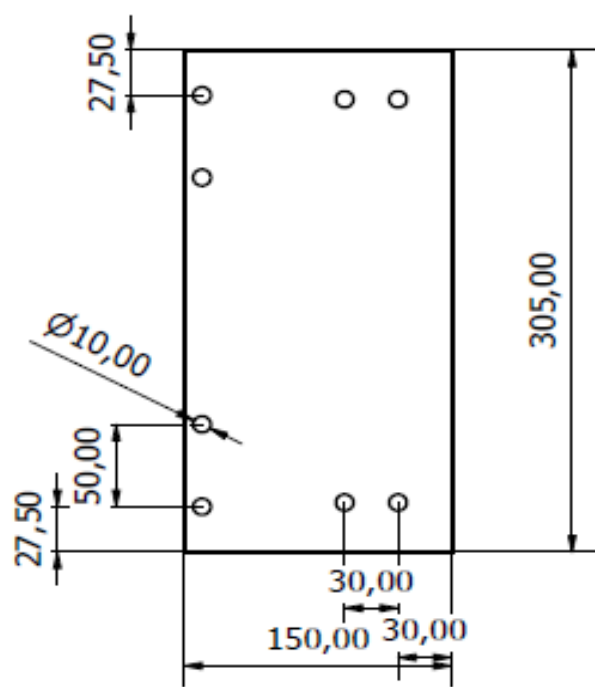
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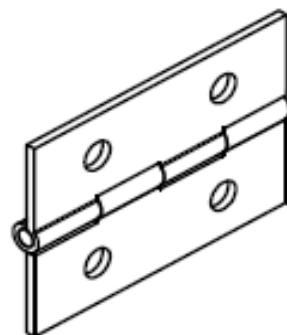
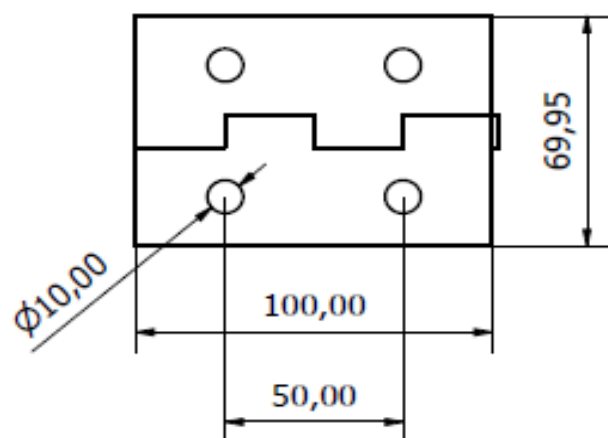
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PROJECT OVERVIEW

The Automatic Stretcher–Wheelchair is an innovative mechanical system designed to assist in the safe and efficient transfer of patients within hospitals, clinics, and ambulances. Traditional patient transportation often requires multiple transfers—from bed to stretcher and then to a wheelchair—which can be physically demanding for healthcare personnel and uncomfortable or even risky for patients. To address this challenge, this project focuses on developing a multifunctional system that can automatically convert from a stretcher to a wheelchair and vice versa at the push of a button.

The system operates using a motor-driven pulley and linkage mechanism that controls the movement of hinged plates forming the backrest, seat, and leg-rest sections. When the motor is activated, the cables pull specific joints, causing the backrest to elevate and the leg-rest to fold down, smoothly transforming the stretcher into a wheelchair. Reversing the motor returns the structure to its flat stretcher form. The frame is built using mild steel for strength and durability, supported by caster and main wheels for mobility and stability. This design allows a single operator to manage patient movement safely and effortlessly.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who have supported and guided us throughout the completion of this project. We are especially thankful to our mentors for their valuable suggestions, encouragement, and feedback at various stages of the work. Their insights have greatly helped us improve the quality of this project. We also acknowledge the use of various resources, references, and tools that provided us with the necessary knowledge base to carry out this work. Finally, we are grateful to each other as members of Group 13 for the teamwork, dedication, and cooperation that made the successful completion of this project possible.

ABSTRACT

This project presents the design and fabrication of an Automatic Stretcher–Wheelchair that merges two essential hospital devices into one efficient, motorized system. The mechanism converts between stretcher and wheelchair configurations using linkages, hinges, and a pulley-cable drive system powered by an electric motor. Unlike conventional designs, the proposed system eliminates the need for sensors or complex electronics, relying instead on a simple mechanical control principle.

During operation, the motor's rotational motion is transmitted through a pulley arrangement, which adjusts the angular positions of the plates to form the seating structure. The transformation process is smooth, stable, and safe for patients. The prototype demonstrates practical feasibility, compact design, and low power consumption. It can be easily operated using a simple switch, making it suitable for hospital corridors, emergency departments, and ambulances. Overall, this project provides a low-cost, user-friendly, and ergonomic solution for improving patient transport systems.

MOTIVATION

The motivation behind developing the Automatic Stretcher–Wheelchair stems from observing the physical strain and time consumption involved in manually transferring patients between beds, stretchers, and wheelchairs. Medical staff often face difficulties in handling patients, especially those who are unconscious, injured, or elderly. Manual lifting not only risks injury to caregivers but also causes discomfort and instability for the patient.

Hence, the project aims to design a mechanically automated system that minimizes manual intervention while ensuring patient comfort and safety. By integrating stretcher and wheelchair functionalities into a single device, this project promotes efficiency, convenience, and innovation in hospital care. The Automatic Stretcher–Wheelchair represents a step toward modernizing patient-handling equipment through simple automation, robust mechanical design, and practical application.

COST ESTIMATION

Cost of Mild Steel used – $14.47 \times 90 = \text{Rs. } 1302$

Cost of Nuts and Bolts used – $0.3 \times 130 = \text{Rs. } 39$

Cost of 3D Printing used – $3 \times 300 = \text{Rs. } 900$

Cost of Lathe Machine used – $3 \times 350 = \text{Rs. } 1050$

Cost of Drilling Machine used – $5 \times 100 = \text{Rs. } 500$

Cost of Electronics Kit used – Rs. 3000

Total Cost = Rs. 6791

